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1.1 Continuous Experimental Apparatus
A flow through extractor designed for use with CO2 under supercritical conditions (Pc = 72.8 atm; Tc = 31°C) and a static batch extractor will be used to complete the above tasks. The work with CO2 will begin in the first year and continue through years 2 and 3. Liquid CO2 delivered from a standard gas cylinder at saturation pressure is compressed by a reciprocating piston pump (Isco). Pressure in the reactor will be controlled with a PI controller. Carbon dioxide flow rate will be controlled with a PI controller adjusting a metering valve. Electronic pressure measurements will be made in the reactor and at suitable points in the process. Thermocouples will be used to measure the CO2 stream temperature at the entrance and exit. Analysis of the extractor effluent will be made by AAand Inductive Coupled Plasma (ICP). A sample collector will also be used in which the CO2 is depressurized and the solute collected in a

Task Description:

suitable solvent. The solute/solvent sample will be AA analyzed and ICP.

1.2 Task 1: Typical samples of components and contaminants.

The samples will come initially from the JCS Mars-1 Martian Regolith Simulate. The samples will be used as received. Since most of the data reporting composition of Martians soils as oxides, we will study the solubility of these oxides. We will also prepare carbonate, sulfate, and hydrated sulfate compound for iron, magnesium, aluminum, and titanium and determine their solubilities in supercritical carbon dioxide.

1.3 TASK 2: SOLUBILITY STUDIES

The solubility in CO2 will be determined at three temperatures, 45°C, 55°C, and 65°C and as a function of pressure in the range between 1500 and 4000 psia. The apparatus and experimental procedure described above will be used.

1.4 TASK 3: CORRELATION OF THE DATA

The Peng-Robinson or Redlich-Kwong equations will be used to correlate the experimental data over the temperature and pressure range studied. The solvent interaction parameters will be determined by regressing the data using standard statistical methods applied to the experimental data.

1.4 TASK 4: PRELIMINARY PROCESS DESIGN

Based on the solubility data, correlations, and removal rates a preliminary design of a pilot scale system will be performed. The preliminary design will include a flowsheet, stream data (temperatures, pressures, compositions, flowrates, densities, etc.), sizing data on the equipment including weight, and energy requirements. The design will be conducted using ASPEN's chemical process simulator. ASPEN's process simulator has extensive capability to regress thermodynamic data. Once the data are entered into the database, they will become the basis of an extended user database which will include all of the data on the over 1700 compounds plus the data from this work. In this form, the data can be easily shared by different investigators. Energy cost to compress the CO2 in the Martian atmosphere from 4 millibar to 75 bar is approximately 0.36 kW/ kg. The objective of this work is the development of a supercritical CO2 process for the recovery of minerals from Martian soils.

Task 1: Samples of the simulant JCS Mars-1 will be obtained from the Johnson space flight center and analyzed by AA. Additional compounds such as metal chlorides, sulfates, and hydrated sulfates, nitrates, carbonates, and oxides will be prepared in mixtures based on the current knowledge of Martian soils.

Task 2: The solubility of minerals and compounds in supercritical CO2 will be determined as a function of temperature and pressure.

Task 3: The solubility data will be modeled using the Peng-Robinson or Redlich-Kwong equations; solvent interaction parameters will be regressed from experimental data. The results of this research will provide the data and proof of the concept for extraction of minerals from the Martian surface. If hydrated forms of minerals are found, it may also be possible to extract water. The process should be able to extract and purify minerals as a first step in processing Martian metals for use in Martian colonization. Bound water may be recovered and oxygen sources may be identified from oxides. The solubility data and behavior under supercritical conditions provide the fundamental basis to determine the feasibility of recovering minerals from Martian soils via supercritical extraction.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task 1
We obtained samples of JCS Mars-1. This simulated Martian regolith was extracted using supercritical CO2. In our experimental procedure to screen samples for solubility in supercritical carbon dioxide, we take about 1 gram of sample, place it in the sample cell of our ISCO extractor. Supercritical carbon dioxide is pumped through the extraction cell at a flowrate of about 1ml/min. The pressure and temperature are controlled. The temperature and pressure range are 35-145 C and 50-240 Bar. Approximately 5.5% was extracted at 35oC. At 120oC, 13% was extracted, as compared to 8% by heating in a furnace. Analysis of this complex material has not yet been performed. Thermal Gravimetric Analysis (TGA) and Differential Scanning Calorimetry (DSC) results for the JCS Mars-1 simulant. indicate a change, either from reaction, vaporization, latent heats of structure rearrangement, or combinations as indicated by a release of 177.3J/g energy as determined by the DSC. We have not been able to determine exactly what materials we extracted. We will continue to look for funding to study the JCS-1 sample via TGA, DSC, XRD, and SEM-EDS to determine the chemical changes occurring.

Task 2

We began to screen inorganic compounds such as metal sulfates, chlorides, nitrates, etc. In our experimental procedure to screen samples for solubility in supercritical carbon dioxide, we take about 1 gram of sample, place it in the sample cell of our ISCO extractor. Supercritical carbon dioxide is pumped through the extraction cell at a flowrate of about 1ml/min. The pressure and temperature are controlled. The temperature and pressure range are 35-145 C and 50-240 Bar. The carbon dioxide exiting the extractor is bubbled through water contained in a glass tube. The aqueous solution in the tube can then be analyzed via ICP for metals. TGA and DSC are also run on these samples to compare the results of heating in a nitrogen environment and to determine the thermal characteristics of the material. The results of the screening show no appreciable extraction for CaCl2, NaI, AgNO3, NH4SO4, K2Cr2O7, (NH4)2MoO4, CuCl2 , and FeCl3 Measureable amounts of material were extracted for the following compounds NH4HF2, CoCl2•6H2O, CuSO4•5H2O, CuCl2•2H2O, Fe(NH4)2•12H2O, Fe(NH4)2•6H2O, FeSO4•7H2O, Fe(NO3)•9H2O, FeCl2•4H2O, MnSO4•5H2O, Ni(NO3)2•6H2O

Our data show that the only compounds which show any significant weight change are those who have associated with them water of hydration. We confirmed, therefore, that it is possible to recover water from hydrated species using supercritical carbon dioxide. Recovering water is significant, since water can be used to support life on Mars directly or the water can be broken down into hydrogen for fuel and oxygen for breathing or as an oxidizer.

We have focused on recovering water of hydration from several of the species tested. We did further testing of FeSO4

Task Progress:	•7H2O to determine the temperatures at which waters of hydration are removed. The first three water molecules are extracted between 50-70 oC. The next three water molecules are extracted between 110-120 oC. To estimate the bond energy, we performed a DSC on a fresh sample of FeSO4•7H2O. The energy required at 75 C to break the bonds which release the first three water molecules is about 145.7 Kj/mol of water. At about 110 C the energy to release the next three waters, approximately 180 kJ/mol. During this step the water is vaporized. The measured latent heat of vaporization of water at 121.5 C, is approximately 39.2 k J/mol. The literature value for the heat of vaporization to be 39.6 kJ/mol, which agrees closely with the experimental data. . A similar study was performed in MgSO4.7H2O. We have devised a method, which allows us to take our extracted samples from the extractor and examine them via DSC without re-exposing the sample to the atmosphere and allowing it to absorb water from the air. At all temperatures there is a enhancement of the water recovered via supercritical CO2 extraction over just direct heating which ranges form 4-9%. Recovery of water via supercritical extraction is possible, but not significantly more than just heating the sample to recover water. To enhance the solubility of metals in supercritical carbon dioxide we have begun using a surfactant. The surfactant we have chosen is high performance perfluoropolyether. This perfluoropolyether has a long nonpolar organic chain which helps make it soluble in the supercritical carbon dioxide and a number of fluorine atoms attached at different points along the chain which help trap the metal species. The overall result of combining supercritical carbon dioxide and the perfluoropolyether surfactant is to enhance the solubility of the metal species. We have examined several magnesium compounds, chloride hexahydrate, MgCl2.6H2O, magnesium sulfate, MgSO4.7H2O, magnesium perchlorate, Mg(ClO4)2 and cupric nitrate trihydrate (Cu(NO3)2.3H2O.. For the cupric nitrate trihydrate, without the surfactant extraction was negligible and was 34% with the surfactant For magnesium chloride hexahydrate extraction was neglible without surfactant and about 32% of the magnesium with the surfactant. Magnesium is a lightweight metal which can be used as fuel in a solid rocket motor. Based on the Mar Pathfinder data reported as MgO, the Martian soil can contain up to 9% MgO. The surfactant significantly enhances metal extraction. We examined the effects of the weight % surfactant, weight % water, and the ratio of %surfactant to % water on the extraction efficiency for magnesium compounds. It appears that about 0.5 weight % of chelate and 0.15 weight % water is required to extract about 30% of the magnesium present in the sample. These extraction experiments are performed by first dissolving the solid in deionized water. The solution is then deposited on a filter paper, dried, and placed in the extraction cell. When the as received materials are placed in the extraction cell without dissolving first the extraction efficiency drops as low as 5%. The difference could be related to particle size. Another possible path to water would be through the reactions of calcium hydroxide or magnesium hydroxide with supercritical carbon dioxide. The carbonate species are favored over the hydroxides in earth environments, but no data exists for carbonate species on Mars. No evidence exists for the presence of hydroxides in the Martian soils although the recent Mars Odyssey data indicate the presence of hydrogen in a chemically or physically bound form. A more likely source of hydrogen would be in species such as Talc, Tremolite (Amphibole), or Chrysotile (Serpentine). The more complex crystal structures are more likely to trap the hydrogen molecules. Water can be released by reaction with supercritical carbon dioxide with these minerals.We have initiated a set of experiments to include the reaction of these species at supercritical carbon dioxide conditions. However, the reactions do not occur measurably below 350 o C. Our extraction apparatus has a limit of 150 OC. We are trying to obtain funding for a 25 cm3 heated reaction vessel which will allow us to study the production of water using this chemistry. Task 3 We have successfully modeled the solubilities of magnesium chloride hexahydrate in SC CO2. We have used standard thermodynamic models which incorporate mixing rules and the Peng-Robinson equation of state to model the copper-chelate complex solubility in supercritical CO2 We obtained the model parameters (Tc, Pc, vs, Psat and k12) by fitting experimental data using the Marquardt-Levenberg least squares (MLLS) method. Good agreement between experimental and calculated results was achieved Our initial efforts to model the solubility of water in supercritical CO2 have been a collaborative effort with Prof. G. Mauer, who has used molecular simulations to estimate these solubilities. The experimental data of Takenouchi and Kennedy report a mole fraction of water in supercritical carbon dioxide of 0.045 while the simulations under predict the solubility at a value of 0.025. . Task 4 A conceptual flowsheet for a process to recover both mineral matter and water from inorganic minerals on Mars has been prepared. As we begin to collect the solubility data we will enter it into a customized database in the ASPEN simulation environment.
Bibliography Type:	Description: (Last Updated: 04/04/2006)
Articles in Peer-reviewed Journals	Wang, T.; Debelak, K.A.; Roth, J.A. "SUPERCRITICAL CARBON DIOXIDE EXTRACTION OF MAGNESIUM(II) CHLORIDE HEXAHYDRATE (MgCl2•6H2O) WITH PFPE AND WATER" submitted Jan 2006, Journal of Supercritical Fluids , Jan-2006
Dissertations and Theses	Wang, Tong. "Supercritcal Extractions of Minerals in Martian Soils." Ph.d Dissertation, December 2005. Dec-2005